Problem 17

Based on the general term of the series, a_n, conclude that the following series diverges:

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n+\alpha}}.$$

Solution

Step 1: Identify the General Term

The general term of our series is:

$$a_n = \frac{1}{\sqrt[n]{n+\alpha}} = \frac{1}{(n+\alpha)^{1/n}}.$$

Step 2: Evaluate the Limit of $(n+\alpha)^{1/n}$

To determine the behavior of a_n as $n \to \infty$, we first need to find:

$$\lim_{n\to\infty} (n+\alpha)^{1/n}.$$

Taking the natural logarithm:

$$\ln\left[(n+\alpha)^{1/n}\right] = \frac{1}{n}\ln(n+\alpha) = \frac{\ln(n+\alpha)}{n}.$$

Step 3: Apply L'Hôpital's Rule

The limit $\lim_{n\to\infty} \frac{\ln(n+\alpha)}{n}$ is of the indeterminate form $\frac{\infty}{\infty}$. Applying L'Hôpital's rule:

$$\lim_{n\to\infty} \frac{\ln(n+\alpha)}{n} = \lim_{n\to\infty} \frac{\frac{d}{dn}[\ln(n+\alpha)]}{\frac{d}{dn}[n]} = \lim_{n\to\infty} \frac{\frac{1}{n+\alpha}}{1} = \lim_{n\to\infty} \frac{1}{n+\alpha} = 0.$$

Therefore:

$$\lim_{n\to\infty} \ln\left[(n+\alpha)^{1/n}\right] = 0.$$

Step 4: Determine the Limit of $(n+\alpha)^{1/n}$

Since the logarithm function is continuous:

$$\lim_{n\to\infty} (n+\alpha)^{1/n} = e^{\lim_{n\to\infty} \ln[(n+\alpha)^{1/n}]} = e^0 = 1.$$

Step 5: Calculate the Limit of a_n

Now we can evaluate:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{(n+\alpha)^{1/n}} = \frac{1}{\lim_{n\to\infty}(n+\alpha)^{1/n}} = \frac{1}{1} = 1.$$

Therefore:

$$\lim_{n\to\infty} a_n = 1 \neq 0.$$

Step 6: Apply the Divergence Test

Since $\lim_{n\to\infty} a_n = 1 \neq 0$, by the Divergence Test, the series:

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n+\alpha}}$$

diverges.

General Observation

This result holds for any real value of α. The key insight is that:

$$\lim_{n\to\infty} b^{1/n} = 1 \quad \text{for any } b > 0.$$

In our case, $b = n + \alpha$, and as long as $n + \alpha > 0$ for sufficiently large n (which is true for any fixed α when n is large enough), we have:

$$\lim_{n\to\infty} (n+\alpha)^{1/n} = 1.$$

Intuitive Explanation

The n-th root of any positive number, no matter how large, approaches 1 as $n \to \infty$. This is because:

$$(n+\alpha)^{1/n} = e^{\frac{\ln(n+\alpha)}{n}},$$

and the ratio $\frac{\ln(n+\alpha)}{n} \to 0$ since logarithmic growth is much slower than linear growth.

Therefore, the terms of our series approach 1, and we are essentially adding 1 infinitely many times, which clearly diverges.

Final Answer

Conclusion

The series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n+\alpha}}$ **diverges** for any real α because:

$$\boxed{\lim_{n\to\infty} \frac{1}{\sqrt[n]{n+\alpha}} = 1 \neq 0.}$$

The fundamental property that $(n+\alpha)^{1/n} \to 1$ as $n \to \infty$ ensures that the general term approaches 1, violating the necessary condition for convergence. By the Divergence Test, the series must diverge.